

```
1
2  #class attribute
3  color='white'
4
5  #1
6  class vehicle():
7      pass
8      def __init__(self):
9          self.color = 'white'
10
11  def vehile(self):
12      print('vehicle color',self.color)
13
14  #2
15  class bus(vehicle):
16      pass
17
18  class car(vehicle):
19      pass
20
21  #3
22  volvo=bus()
23  volvo.color
24
25  tata=car()
26  tata.color
27
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
'white'
>>> tata=car()
>>> tata.color
'white'
```

```
29
30 #4
31 class country:
32     country='usa'
33
34 #5
35 class bank:
36     currency = 'USD'
37
38 class SavingsAccount(bank):
39     currency = 'Pounds'
40
41 class CheckingAccount(bank):
42     currency = 'Rupees'
43
44 ch=CheckingAccount()
45 ch.currency = 'euro'
46 ch.currency
47
48
49
50 #6
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
>>> ch=CheckingAccount()
>>> ch.currency = 'euro'
>>> ch.currency
'euro'
```

```

49
50 #6
51 class car:
52     def __init__(self,__engine_number):
53         #private attribute can't overwrite
54         self.__engine_number = __engine_number
55
56     def get_engine_number(self):
57         return self.__engine_number
58
59 c1=car(26)
60
61 c1.__engine_number = 33 #can't overwrite
62
63 c1.get_engine_number()
64
65
66
67 #7

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\prem2\Git files\task-code\py> C:/Users/prem2/AppData/L
...         return self.__engine_number
...
>>> c1=car(26)
>>> c1.__engine_number = 33 #can't overwrite
>>> c1.get_engine_number()
26

```

```

67  #7
68  from abc import ABC, abstractmethod
69  class Animal(ABC):
70      |
71      | @abstractmethod
72      | def speak(self):
73      |     pass
74
75  class Dog(Animal):
76      | def speak(self):
77      |     return 'boww'    #implementation
78
79  class Cat(Animal):
80      | def speak(self):
81      |     return 'meoww'   #implementation
82
83  d1=Dog()
84  d1.speak()
85
86  c1=Cat()
87  c1.speak()
88

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\prem2\Git files\task-code\py> C:/Users/prem2/AppData/L

```

>>> d1=Dog()
>>> d1.speak()
'boww'
>>> c1=Cat()
>>> c1.speak()
'meoww'

```

```

90  #8
91  class vehicle():
92      def __init__(self,max_speed,mileage):
93          self.max_speed = max_speed
94          self.mileage = mileage
95
96      def details(self):
97          print('speed:', self.max_speed)
98          print('mileage:', self.mileage)
99
100  car = vehicle(210,15)
101  car.details()
102
103
104
105
106  #9
107  class add:
108      def __init__(self,num):
109          self.num = str(num)
110

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\prem2\Git files\task-code\py> C:/Users/prem2/App
...         print('mileage:', self.mileage)
...
>>> car = vehicle(210,15)
>>> car.details()
speed: 210
mileage: 15

```

```
106 #9
107 class add:
108     def __init__(self,num):
109         self.num = str(num)
110
111     def calculate(self):
112
113         total = 0
114
115         for i in self.num:
116             total = total+(int(i)**2)
117
118         return total
119
120 number = int(input('enter the 3-digit number:'))
121
122 o1 = add(number)
123 print(o1.calculate())
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\prem2\Git files\task-code\py> C:/Users/prem2/AppData

...

>>> number = int(input('enter the 3-digit number:'))

enter the 3-digit number:345

>>> o1 = add(number)

>>> print(o1.calculate())

50